European Review of Agricultural Economics Vol **50 (1)** (2023) pp. 1–28
doi:https://doi.org/10.1093/erae/jbac023
Advance Access Publication 6 October 2022

Regression discontinuity designs in agricultural and environmental economics

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Received May 2022; final version accepted September 2022

Abstract

Regression discontinuity designs (RDD) are increasingly being employed in agricultural and environmental economics to identify causal effects. Here, we showcase recent applications, identify best practices, discuss commonly invoked identifying assumptions and show how these can be tested. We discuss basic empirical issues and more advanced topics, including how to exploit the availability of panel data, models to explain heterogeneous treatment effects and extrapolation of local estimates. Moreover, we show how agricultural economists can leverage RDD in combination with remote sensing and environmental modelling. Finally, we highlight three areas of emerging opportunities and draw conclusions for research and policy.

Keywords: regression discontinuity design, difference-in-discontinuities, causal inference

JEL classification: C01, Q00

1. Introduction

The identification of causal effects takes centre stage in agricultural economics and policy and other fields of applied economics (Esposti and Sotte, 2013). It is the basis for evidence-based policy recommendations. Since the 'credibility revolution' in applied economics, researchers increasingly often either design their own experiments, for which they randomise treatment and ensure there is a valid control group, or they use a research design that allows for causal inference with non-experimental data (Angrist and Pischke, 2010). Among such research designs, the regression discontinuity design (RDD) is among the most popular (Abadie and Cattaneo, 2018; Brodeur, Cook and Heyes, 2020). Compared to other non-experimental research designs for causal

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inference (such as difference-in-differences or instrumental variables), RDDs offer the key advantages that the identifying assumptions can be especially transparently tested and arguments can be visibly supported by informative plots of the data. As a consequence, publications based on RDDs are less affected by p-hacking and publication bias compared to alternative research designs (Brodeur, Cook and Heyes, 2020). Comparisons of randomised control trials and RDDs support the general notion that they tend to be similarly reliable to identify causal effects (Chaplin *et al.*, 2018; Gleason, Resch and Berk, 2018).

The main drawback of the design is a potentially low external validity because causal identification is highly local. It should be noted, however, that there exists an array of technical solutions to address this concern (Wing and Bello-Gomez, 2018). More generally, while clearly not all policy questions related to agricultural and environmental issues can be answered using RDDs, it is a highly valuable tool in the empirical toolkit of agricultural and environmental economists, and applied policy analysis and its potential has by far not been fully exploited yet.

We here aim to showcase applications in agricultural and environmental economics, discuss practical empirical issues along with best practices and explain recent methodological insights.[1] These applications mostly focus on the intersections of agricultural and development economics (e.g. the effect of new roads on rural development) and agricultural and environmental economics (e.g. the effect of a development policy on forest cover).[2] Finally, we discuss recent advances and extensions, and present an outlook on areas in which we see particular promise for future research in agricultural and environmental economics and policy.

1 For general overviews of the method, we refer the interested reader to Cattaneo, Idrobo and Titiunik (2020). *A Practical Introduction to Regression Discontinuity Designs: Foundations and Extensions*. Cambridge University Press, Cattaneo and Titiunik (2021). 'Regression Discontinuity Designs.' *arXiv preprint arXiv:2108.09400*, Imbens and Lemieux (2008). 'Regression Discontinuity Designs: A Guide to Practice.' *Journal of econometrics*, 142(2), 615–35, Lee and Lemieux (2010). 'Regression Discontinuity Designs in Economics.' *Journal of economic literature*, 48(2), 281–355. In addition, Cunningham (2021). *Causal Inference*. Yale University Press. and Huntington-Klein (2021). *The Effect: An Introduction to Research Design and Causality*. Chapman and Hall/CRC. Provide great introductory treatments including codes and exercises in Stata, Python and R.

2 Note that we omit studies here from environmental and development economics that do not strongly overlap with agricultural economics. For example, Asher and Novosad (2017). 'Politics and Local Economic Growth: Evidence from India.' *American Economic Journal: Applied Economics*, 9(1), 229–73, Bagues and Campa (2021). 'Can Gender Quotas in Candidate Lists Empower Women? Evidence from a Regression Discontinuity Design.' *Journal of Public Economics*, 194, 104,315, Chen, Shi and Tang (2019). 'Valuing the Urban Hukou in China: Evidence from a Regression Discontinuity Design for Housing Prices.' *Journal of Development Economics*, 141, 102,381, Filippini and Wekhof (2021). 'The Effect of Culture on Energy Efficient Vehicle Ownership.' *Journal of Environmental Economics and Management*, 105, 102,400, Khanna and Zimmermann (2017). 'Guns and Butter? Fighting Violence with the Promise of Development.' *Journal of Development Economics*, 124, 120–41, MacPherson and Sterck (2021). 'Empowering Refugees through Cash and Agriculture: A Regression Discontinuity Design.' Ibid.149, 102,614, Salman *et al.* (2022). 'Paris Climate Agreement and Global Environmental Efficiency: New Evidence from Fuzzy Regression Discontinuity Design.' *Energy Policy*, 168, 113,128, Shi (2022). 'Land Reallocation Responses to China's New Rural Pension Scheme: Evidence from a Regression Discontinuity Design.' *Journal of Rural Studies*, 92, 35–49.

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There are several particularities of agricultural, environmental and related fields in applied economics that open up interesting possibilities to use some version of a RDD, e.g. data structure and availability as well as policy design and implementation. As we discuss in more detail below, the most important requirement to apply some version of RDD is the existence of a continuously distributed variable (often called the 'running variable') that contains a clearly defined cutoff that affects which units in the data are assigned to 'treatment' and which to 'control'. Under a list of well-understood and falsifiable assumptions, this set-up then follows a generally similar logic to an actual experiment. Especially in agricultural contexts, programs and policies are often implemented such that we can find these running variables with thresholds. For example, agricultural extension services might set boundaries such that only farms within a certain radius receive services (Pan, Smith and Sulaiman, 2018) and sometimes governments use arbitrary population, poverty, or farm size thresholds (Alix-Garcia *et al.*, 2013; Asher and Novosad, 2020; Chang and Lin, 2016) as basis for infrastructure investments, transfers, or regulations (Ayres *et al.*, 2021; Jones *et al.*, 2022).

Data availability is another key aspect because the farming sector is often particularly well quantified. In Europe, for example, census and bookkeeping data is available as panel data allowing often to exploit developments over time and space (e.g. Finger and El Benni, 2021). Moreover, remote sensing and computational capabilities offer exciting opportunities, because many studied phenomena can be measured from space (e.g. cropland expansion or forest loss) (Hansen *et al.*, 2013; Potapov *et al.*, 2021). Remote sensing data offers many advantages, such as often seamless global coverage and many repeated measurements (Burke *et al.*, 2021; Lobell *et al.*, 2019). The available data can furthermore be extended by using environmental modelling, which is helpful to obtain data on phenomena that can only partly be observed using satellites, such as land degradation (Wuepper *et al.*, 2021). Together, the large and growing amount of available data alongside the existence of many useful thresholds that allocate units into 'treatment' and 'control' is a boon to the use of RDDs agricultural economics and policy and related fields.

The remainder of this review is structured as follows: In Section 2, the fundamentals of RDD are discussed. First, we lay out the fundamental logic of the research design and showcase exemplary applications (2.1), followed by a stylised 'recipe' which shows practical empirical issues one may encounter and what to do about them (2.2). Section 3 covers advanced topics. It begins with a discussion of the difference-in-discontinuities design (3.1), models for heterogeneous treatment effects (3.2), alternatives to using linear regression (3.3), approaches to learn about effects away from the cutoff (3.4), how to quantify spillovers (3.5) and combining the design with agricultural and environmental modelling (3.6). Section 4 is about emerging (data) opportunities, we emphasise the growing availability of high-resolution geospatial measurements (#1), data generated with machine learning (#2) and data that contains policy relevant thresholds that have not been exploited yet (#3). Section 5 concludes.

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2. Fundamentals of regression discontinuity design

In this section, we begin with a brief overview of the method (2.1), followed by a showcasing of a few empirical applications (2.2), a list of empirical issues and best practices how to handle them (2.3).

2.1. The basic logic and exemplary applications

The characteristic feature of every RDD is that a particular situation is required: Units (e.g. farms or fields) can be linearly sorted along a 'running variable', e.g. farm size, income or farms' distance to a specific point in space. The treatment is assigned on the basis of a sharp cutoff value, e.g. every farm below a certain size is eligible for a certain program, or a certain regulation only applies to fields that lie west of a certain border. Under clearly defined conditions, this set-up provides a natural experiment, with some similarities to an actual randomised experiment: The units above the cutoff are in the 'treatment group', the units below the cutoff are in the 'control group' (Lee and Lemieux, 2010). Causal identification of treatment effects comes from the discontinuous change in the probability of receiving the treatment whereas all potential confounders are either continuously changing across the cutoff, or do not differ just above and below the cutoff (Cattaneo and Titiunik, 2021).

Formally, let treatment assignment be denoted by $D \in \{0,1\}$, the running variable by X and the cutoff by c. Then, there is a mechanism that $D = 1$ if $X \geq c$ and $D = 0$ if $X < c$. The regression, in its most basic form, looks as follows:

$$Y = \alpha + \tau D + \beta_1 X^{above} + \beta_2 X^{below} + \varepsilon \tag{1}$$

Y is the outcome of interest (e.g. farmers income), α is a constant, β_1 and β_2 smoothly control for the distance to the cutoff from above and from below (with X e.g. being farm size), τ is the estimated discontinuity and ε is an error term. This regression is then estimated on a subsample of the data that is statistically optimally close to the cutoff value. The reason why the entire available data is not used (i.e. further beyond the cutoff value) is a trade-off between variance and bias (Calonico, Cattaneo and Farrell, 2020; Cattaneo and Vazquez-Bare, 2016). To reduce the influence of confounding factors, we want to restrict the data as close as possible to the cutoff where units are most comparable to each. However, this reduces the number of available observations and thus makes our estimates and causal inference increasingly imprecise, limiting the ability to assess policies. Thus, one needs to identify an optimal 'bandwidth' around the cutoff that optimally balances variance and bias[3]

3 https://rdpackages.github.io/rdrobust/ provides a repository of software to perform all RDD specific tasks, such as for finding the optimal bandwidth, calculate the statistical power of a given design and many more.

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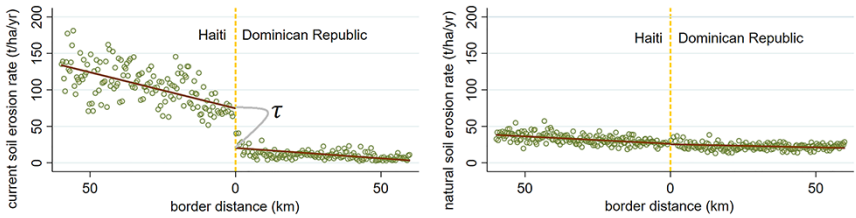

Fig. 1. Discontinuity plots for the actual and the natural rate of soil erosion at the border of Haiti and the Dominican Republic. Data from Wuepper, Borrelli and Finger (2020a).

(Calonico, Cattaneo and Farrell, 2020; Cattaneo and Vazquez-Bare, 2016). The effect of interest is then identified by the following:

$$\tau = \mathbb{E}\left[Y(1) - Y(0)\,|X = c\right] = \lim_{x\downarrow c} \mathbb{E}\left[Y|X = x\right] - \lim_{x\uparrow c} \mathbb{E}\left[Y|X = x\right] \quad (2)$$

The first part simply defines that at the cutoff c the causal effect is defined as the difference between the factual and the counterfactual $Y(1) - Y(0)$. The second part specifies that this is given by the vertical distance at the cutoff between the two conditional expectations, $\lim_{x\downarrow c} \mathbb{E}\left[Y|X = x\right] - \lim_{x\uparrow c} \mathbb{E}\left[Y|X = x\right]$.

A much appreciated feature of the RDD is that the identification mechanism can be visually inspected in the raw data. As defined above, the causal effect is given by the vertical distance at the cutoff between the two conditional expectation functions above and below the cutoff and it is commonly expected that this can be seen when the data is plotted, as in Figure 1 from Wuepper, Borrelli and Finger (2020a).

They estimate whether there are sharp discontinuities in soil erosion at international borders that are caused by political and economic differences between the divided countries. An illustrative example in Figure 1 is provided by the border between Haiti and the Dominican Republic. A particularity of the study of Wuepper, Borrelli and Finger (2020a) is that they do not only analyse the current rate of soil erosion globally (the factual, left panel of Figure 1) but also explicitly model and then analyse the natural rate of soil erosion without human activity (the counterfactual, right panel of Figure 1). This comparison nicely illustrates the basic logic of the RDD. The plots show soil erosion on the Y-axis and border distance on the X-axis, the circles show local soil erosion averages for different border distances and the fitted regression lines indicate the spatial trend in soil erosion, on each side of the border. It can be seen that both currently and also naturally, Haiti has a higher rate of soil erosion, mainly because rainfall increases from the side of the Dominican Republic to the side of Haiti (right panel, Figure 1). Yet, naturally, there would not be a discontinuity at the country-border, because factors such as rainfall are spatially smoothly distributed. However, we observe that the rate of soil erosion sharply jumps when one crosses the border from one country to the other

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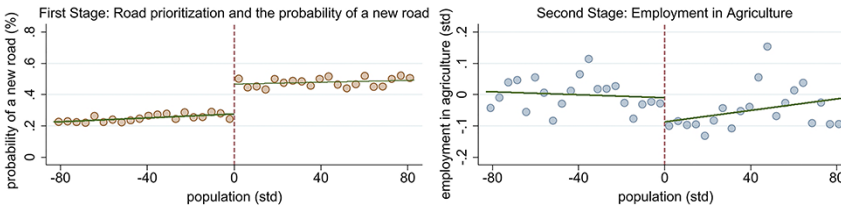

Fig. 2. Discontinuity plots for the effect of road prioritisation, based on population thresholds, on the probability of road construction in India (first stage, left) and the effect of those roads on local employment in agriculture (second stage, right). Data from Asher and Novosad (2020).

(left panel, Figure 1), and this is because of a stark contrast in land cover. This is an example of a spatial RDD because the running variable is a spatial distance.

The study by Asher and Novosad (2020) provides an example of a different kind of RDD, in two aspects (Figure 2). First, their RDD is 'fuzzy' and not 'sharp', meaning that only the *probability* of treatment discontinuously changes at the cutoff. Second, their running variable is not a distance, but it is rural populations. They study the effect of road construction on rural economic development expressed as agricultural employment in India. They exploit that the Indian government aimed to connect as many previously unconnected people at the lowest cost, so they prioritised larger villages over smaller ones and used arbitrary population thresholds from a national census to choose the locations for the new roads. The left plot of Figure 2 shows how on average, the probability of a new road discontinuously doubles right at these thresholds. The right plot then shows one of their several outcomes, which is an index of employment in agriculture. Right at the average population threshold, the new roads enabled labour to leave agriculture and thus, we see a sharp drop in agricultural employment.

Commonly, the most important aspect of RDDs is that a discontinuity in the outcome(s) of interest can only be explained by the discontinuity in treatment assignment (so in the sharp version, only the treatment is discontinuously distributed at the cutoff, and in the fuzzy version, only the probability of the treatment is discontinuously distributed at the cutoff). Also here, it helps to think of the design like of an actual experiment. Treatment and control groups might only differ because of the treatment, not because of any other factors. Unlike with an actual experiment (Behaghel, Macours and Subervie, 2019; Colen *et al.*, 2016), however, we are exploiting a naturally arisen situation and thus we need to be very careful whether we have, indeed, only a discontinuity in treatment assignment, while all potential confounding factors are either continuously or randomly distributed across the threshold.[4] First of all,

4 These two assumptions lead to two distinct frameworks: The continuity framework and the local randomisation framework (Cattaneo and Titiunik, 2021). We discuss examples for both frameworks in Section 2.

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this requires that given contextual information and economic theory, this is to be expected or at least not unlikely. Secondly, there is a list of testable implications. In empirical applications, it is commonly expected that there is sufficient context and data background to support a convincing explanation why the assumptions of the chosen RDD are plausibly fulfilled and the list of identifying assumption is mirrored by a list of supporting, robustness and falsification tests.

As discussed above, the main requirement for the RDD is the existence of a running variable with a clearly defined cutoff value at which treatment assignment jumps—but potential confounding factors do not. While this sounds restrictive at first, there is a perhaps surprisingly large number of empirical policy contexts in which such a mechanism can be found. For example, Alix-Garcia *et al*. (2013) estimate the effect of a large-scale poverty alleviation program in Mexico on deforestation induced by agricultural land expansion. The program was rolled out according to sharp eligibility thresholds in a 'marginality index', such that communities 'just poor enough' received a large income shock, whereas communities 'just slightly too well off' did not. Thus, the analysis is based on a comparison of almost similarly poor rural communities, some of which just had an income-boost whereas others did not. In Sweden, Nordin and Lovén (2020) estimate the effect of a subsidy paid to young farm managers, considering the probability of farm's generational transition, farm income, off-farm income and farm survival. They exploit the program's eligibility cutoff at exactly age 40, such that their analysis is based on farmers just young enough (age ≤ 40) and farmers who just missed this cutoff (age > 40). Ali, Deininger and Goldstein (2014) estimate the impact of

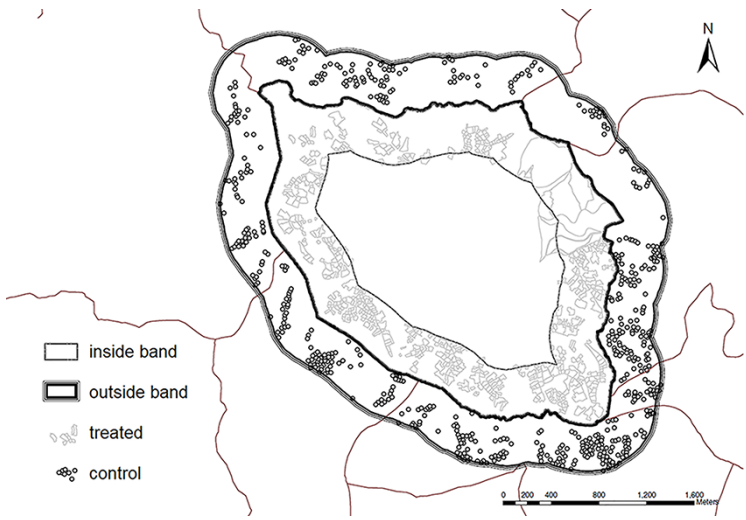

Fig. 3. The sampling strategy of Ali, Deininger and Goldstein (2014). Adapted from Ali, Deininger and Goldstein (2014).

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a land registration pilot in Rwanda on farmers' investments in soil conservation. They administered their own survey among farmers just inside just and outside the regions where the program was piloted and use distance to the pilot-region's border as their running variable (Figure 3 shows the example of one of their sampling cells).

Also historical events are frequently found to provide discontinuities that are useful for RDDs. Appau *et al*. (2021) for example, estimate the effect of bombings and use of Agent Orange during the Vietnam War on current agricultural productivity in the country, using distance to the border between North and South Vietnam as the running variable. In Germany, both Wuepper, Wimmer and Sauer (2020c) and Noack *et al*. (2022) use the historical division of Germany into East and West as a natural experiment to study the environmental impact of different farming structures. Under Communism, the farms in East Germany were expropriated and collectivised. The legacy of this persists today, such that the farms on the western side of the border tend to be small and the farms on the eastern side large (Figure 4). This allows the identification of the effect of farming structures such as are small farms more sustainable (Wuepper, Wimmer and Sauer, 2020c) and do more heterogeneous agricultural structures result in more diverse bird populations? (Noack *et al.*, 2022).

2.2. Practical empirical issues

RDDs are generally considered a particularly credible way to identify causal effects, mainly because all identifying assumptions are transparently falsifiable. However, the credibility of a specific application depends on the specific

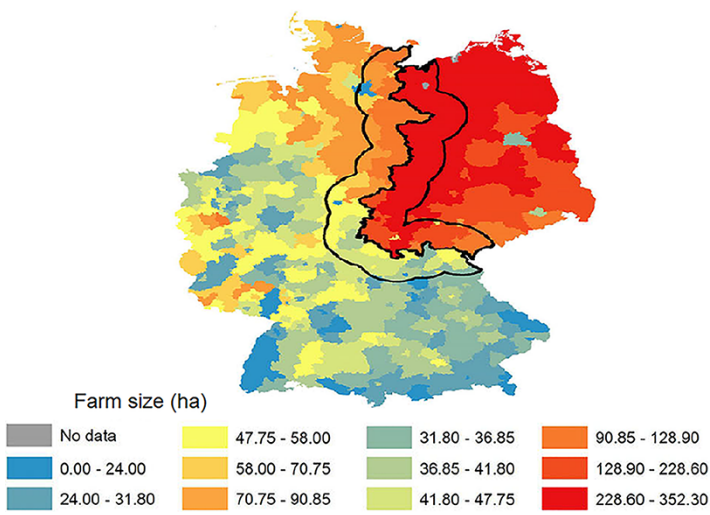

Fig. 4. Germany's historical 'iron curtain' and its farm sizes. Adapted from Noack *et al*. (2022).

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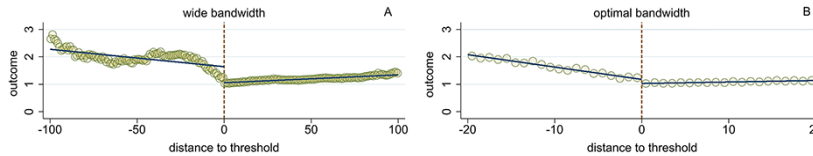

Fig. 5. Non-linear distributions can be misleading, especially in wider bandwidths. Data simulated.

fit and execution of the approach. Here, we present a list of empirical issues and discuss best practices for handling them.

2.2.1. #1 Misleading distributions

It is common to begin regression discontinuity studies with plotting the data as shown in Figures 1 and 2. An important motivation for this is to understand how 'obvious' the discontinuity is in the raw data. There are different kinds of 'misleading' distributions that can lead researchers to confuse a continuous non-linear function with a discontinuous linear function. Figure 5 shows simulated data without a statistically significant discontinuity at the threshold. The circles are small local averages of the outcome and the fitted lines are from a separate regression fitted on either side of the threshold. Figure 5a shows an example in which the bandwidth is set too wide and data points relatively far away from the threshold change the slope of the regression lines such that an artificial discontinuity is created. This potential bias is one reason why it is generally important to present a discontinuity plot in addition to the econometric estimates. Figure 5 shows that when an exemplary statistically optimal bandwidth is being used, the issue is solved.

It is also common in the literature to not only linearly control for distance to the threshold but to also include higher order polynomials as a robustness check. For example, Salazar *et al.* (2020) aim to estimate the impact of an integrated pest management program in Peru to control fruit fly infestation, that was implemented within a clearly delineated geographic area and not outside of it. In their main specifications, they include linear controls for border distance, as suggested by Gelman and Imbens (2019), but as a robustness test, they also include squared terms. Another solutions is to give decreasing weight to units with increasing distance to the threshold. Pan, Smith and Sulaiman (2018) for example, estimate the effect of an extension program in Uganda that only provides service to farms within a 6 km radius and not beyond that. In their specifications, they use a non-parametric regression with a triangular kernel. Bonilla-Mejía and Higuera-Mendieta (2019) use this approach to estimate the effect of protected areas in Colombia on deforestation, using the border of the protected areas as threshold.

Finally, there is an important role for placebo tests. Especially important to show is that by randomly moving the threshold, one does not find similar discontinuities to the one at the actual threshold. In the study by Noack

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et al. (2022), the threshold is the historical inner-German border which separated East and West Germany and continues to mark the divide between mostly large-scale and mostly small-scale farming operations. They are interested in the effect on bird populations and they use two data sources. As a placebo test, they move the threshold stepwise away from its actual coordinates (e.g. 5 km away from the actual border). In one of their datasets, they find that moving away from the actual border turns most estimates statistically insignificant. However, in their other dataset, they find many and often even larger, discontinuities where none should be. Thus, they conclude that their two datasets are not equally reliable for their analysis. Also Ruan, Cai and Jin (2021) use placebo analyses to test the reliability of their estimates of the effect of COVID-19 lockdowns on vegetable prices in China. First, they repeat their analysis in data from before COVID-19 (2018 and 2019) and second, they move the timing of the lockdowns in their data to just three weeks before and three weeks after the actual lockdowns. These placebo tests return statistically insignificant and small in magnitude estimates.

2.2.2. #2 Endogenous thresholds

The most fundamental assumption of the RDD is that the threshold is exogenous. If the threshold is endogenous there is no causal identification. To probe the exogeneity of the threshold, there are two common strategies that are not mutually exclusive. The first strategy is to replace the outcome variable with pre-treatment covariates and see whether there was already a discontinuity in one of these potential confounding variables. For example, Alix-Garcia *et al*. (2018) estimate the effect of a large agri-environmental scheme in Mexico on the adoption of sustainable land management practices and on social capital. They exploit the fact that the Mexican government assigned all applicants a score and used multiple sharp eligibility cutoffs for who was then rejected and who was accepted. The cutoffs were arbitrarily set such that the best applicants got funding and each state used up its budget. However, to be sure of the cutoffs' exogeneity, Alix-Garcia *et al*. (2018) also show that there was no discontinuity in transport infrastructure, distance to major cities, members with rights to communal land, poverty index, topography, canopy cover in 2000 (prior to the program), education and agricultural structures.

The second strategy has, e.g. recently been showcased by Bonilla-Mejía and Higuera-Mendieta (2019), who estimate the effect of Colombia's protected areas by using their boundaries as threshold. To test whether environmental covariates change at these boundaries—individually and jointly—they use the test developed by Canay and Kamat (2018)[5]. Importantly, if a threshold is found to be endogenous, the design cannot be 'rescued' by including 'control' variables (see e.g. Cattaneo and Titiunik, 2021 for a discussion). Exogenous covariates mainly have the following functions: to explore the plausibility that

5 This can be done in Stata or R using rdperm and RATest, respectively: http://sites.northwestern.edu/iac879/software/.

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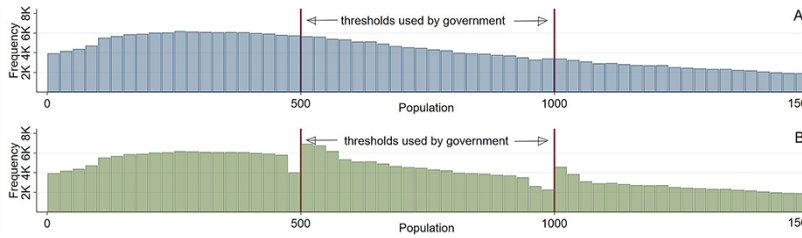

Fig. 6. Bunching next to the thresholds indicates endogenous sorting. Distribution A shows no obvious bunching but distribution B does. Data for A from Asher and Novosad (2020), B is simulated, i.e. hypothetical data.

the threshold is exogenous, to create interesting subsamples from the data, and to potentially make the estimates more precise.

2.2.3. #3 Endogenous sorting

Sometimes, the concern is not that the threshold is endogenous (as discussed above) but that who ends up on which side of the threshold is endogenous. Economic theory is based on the idea that people respond to incentives, so if only farms below 10 hectares can receive a subsidy, or only farmers earning less than 50,000 USD per year, some farmers might respond to this and move just across the threshold to the subsidy-receiving side. If at least some farmers choose themselves whether they are in the 'treatment' or the 'control' group, this invalidates the research design. McCrary (2008) suggests a simple test for this.[6] The fundamental idea is that if sorting happened, we would expect to see a bunching of observations just across the threshold.

An example is provided by the already mentioned study of Asher and Novosad (2020). They estimate the effect of new roads on rural development in India and exploit that the Indian government used haphazard population thresholds to prioritise more densely population regions. To explore whether there might have been endogenous sorting, they look for any suspicious bunching near the used population thresholds. Figure 6 shows their data (Figure 6A), which exhibits no bunching, and simulated data, just for illustration, which does exhibit bunching (Figure 6B).

2.2.4. #4 'Missing' units

The argumentation that units just above and below the treatment threshold are comparable to each other requires that there is enough data just above and below the threshold and that most units have a mirror-unit on the other side of the threshold. For example, Salazar *et al.* (2020) estimate the effects of a fruit fly eradication program that only served farmers within a particular geographic area in Peru. Because they carefully excluded those areas where the

6 Which can e.g. be conducted using 'rddensity', available from https://rdpackages.github.io/rdrobust/.

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treatment boundary coincided with administrative boundaries, Salazar *et al.* (2020) had to ensure that they actually have a sufficient number of villages sufficiently close to the boundary, such that they do not estimate a discontinuity only between far away villages or between a very small number of villages.[7] Their running variable is the distance to the closest neighbour on the opposite side of the border and they provide multiple plots to show the sample distributions in their raw data. In addition, placebo tests are highly informative, because the fact that potential confounding factors show no discontinuities is a good sign that the villages are in fact comparable to each other. These checks are especially important in geographic RDDs, as in a three dimensional setting, even two units with the same distance to the threshold could possibly be far away from each other (e.g. two units could be right on the opposite side of border, but one all the way in the North and one all the way in the South) (Keele and Titiunik, 2014).

2.2.5. #5 Measurement error

Both random and systematic measurement errors can be important in regression discontinuity analyses. Often, random measurement error will only make the estimates less precise, whereas systematic measurement error can—in the worst case—create or remove discontinuities all together. In their analysis of countries' role in the global relationship between crop yield gaps and nitrogen pollution, Wuepper *et al.* (2020b) use international borders as the thresholds at which the influence of countries abruptly changes, whereas they argue that confounders change continuously. Regarding random measurement error, it is clear that globally modelled crop yields and nitrogen pollution only correlate with actual crop yields and nitrogen pollution. Regarding systematic measurement error, the data is on average better for high-income countries than for low-income countries. To explore the implications of both kinds of measurement error, Wuepper *et al.* (2020b) perform a subsample-analysis that excludes all pixels for which only low-resolution data is available and find the same average discontinuities as before.

2.2.6. #6 Imperfect 'compliance'

Like in randomised experiments, units do not always perfectly comply with their treatment assignment. Also like in randomised experiments, one solution to this is to estimate 'intention to treat' effects and another solution is to estimate a two-stage instrumental variables regression with treatment assignment as the instrument (Angrist, Imbens and Rubin, 1996). In a RDD, the treatment assignment is the threshold, so one simply estimates two stages least squares (2SLS) with the threshold as instrumental variable. This is called the 'fuzzy' RDD, because the threshold only increases the probability of treatment, in contrast to the 'sharp' RDD, in which treatment is determined by the threshold. For example, De Janvry, Ritchie and Sadoulet (2016) are interested in the

7 To calculate statistical power, sample size and minimum detectable effects, 'rdpower' can be used, available from https://rdpackages.github.io/rdrobust/.

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effects of a large-scale weather insurance scheme in Mexico. They exploit that the insurance is triggered by a threshold amount of rainfall, such that there are farmers who are exposed to very similar amounts of rainfall, but some of them are right above and some are right below the threshold. Even though the rule is clear, treatment in their study is not deterministic. Being below the threshold only increases the probability to receive an insurance payout. Thus, they use the rainfall threshold as an instrumental variable in a 2SLS framework, with % of the farming area with actual insurance payout as treatment variable. Another example for this design is provided by Ge, Edwards and Akhundjanov (2020) who estimate the effect of property rights on agricultural productivity, focusing on the border of tribal trust land in Utah. As an instrument for today's tribal trust land, they use the boundaries from a historical land allotment.

2.2.7. #7 External validity

External validity can be limited in the RDD. Because the entire identification strategy is based on a local identification of causal effects the natural question is how generalisable are those estimated effects for units away from the threshold? Simple strategies to explore this question are to plot the raw data for a much wider bandwidth than what is used in the analyses to eyeball how particular the threshold area is (Alix-Garcia *et al.*, 2018; Burgess, Costa and Olken, 2018) and to regress the data used for the analysis against the average population (Wuepper, Borrelli and Finger, 2020a). Another possibility is demonstrated by Nguyen and Rieger (2017) who estimate the effect of a community-driven development program in Morocco that was rolled out only in communities with a poverty rate of 30 per cent and above. To explore external validity, they plot how their estimates change in response to enlarging the bandwidth beyond what was estimated as optimal. The more generalisable the estimate, the less it should be affected by this. Rodriguez *et al.*, 2022) perform a test proposed by Dong and Lewbel (2015), they estimate the local income effect of Colombia's coffee farming World heritage site by using the fact that its border was drawn right through the traditional coffee growing region of Colombia. The test then consists of probing whether the effect would have been estimated similarly if the threshold was placed at slightly different values of the running variable. This is based on the derivative of the treatment effect with respect to the running variable (Rodriguez *et al.*, 2022).

3. Extensions of the standard regression discontinuity design

In this section, we present extensions of the standard RDD. We begin with a discussion of the difference-in-discontinuities design, which is a combination of difference-in-differences and RDD (3.1). Next, we present models for heterogeneous treatment effects, which include quantile regression and machine learning models (3.2), followed by alternatives to using linear regression (3.3), approaches to learn about effects away from the cutoff (3.4), how to quantify

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spillovers (3.5), and finally combining regression discontinuity analyses with agricultural and environmental modelling (3.6).

3.1. Difference-in-discontinuities

In agricultural and environmental economics, we can often rely on panel or repeated cross-sectional data. In this dynamic setting the difference-in-discontinuities design can be used to evaluate policies. For example, to identify the forest impact of a place-based economic policy in India (Garg and Shenoy, 2021) or assessing the effect of a strengthening of indigenous property rights in Brazil on deforestation (Baragwanath and Bayi, 2020), or the effect of within-country cultural differences on how farmers respond to a national increase in agri-environmental payments (Wang *et al.*, 2022). The design closely resembles a difference-in-differences set-up and shares the same identifying assumptions with difference-in-differences and RDDs[8] (Butts, 2021; Shenoy, 2018). The main additional assumption on top of what we cover in Section 2.2 is the parallel trends assumption from difference-in-differences, which is however, much weaker when focusing on borders than when comparing the dynamics of units further apart. For example, Shenoy (2018) estimates the regional development effect of a place-based policy in India, using the border of a newly created state for identification. He shows that parallel trends assumption holds near the border but not when using data from too far away. Thus, ceteris paribus, the differences-in-discontinuities can have a higher internal validity than difference-in-differences. The design also solves a common identification issue that plagues the cross-sectional RDD, especially the spatial version: compound treatments (Keele and Titiunik, 2014; Butts, 2021). Consider the study of Garg and Shenoy (2021) who analyse whether a place-based policy that improved economic development[9] had a negative side effect on deforestation. Cross-sectionally, one would be worried that an estimated discontinuity in deforestation after the policy was implemented already existed before the policy was implemented. Many variables differ between the states of India, and many could affect deforestation. However, using differences-in-discontinuities, Garg and Shenoy (2021) show that before the policy was implemented, economic development (measured using nighttime luminosity) and deforestation developed in parallel near the border and after the policy was implemented, economic development diverged (left), but tree cover did not (right) (Figure 7).

8 Also whether an estimate is an average treatment effect, a local average treatment effect, a treatment effect on the treated, or something else, depends—like in the difference-in-difference design—on the context of the study.

9 Shenoy (2018). 'Regional development through place-based policies: evidence from a spatial discontinuity.' *Journal of Development Economics*, 130, 173–89.

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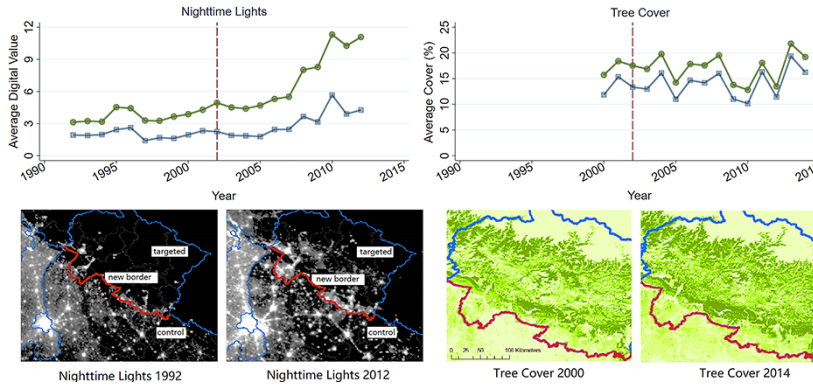

Fig. 7. Top: Change in nighttime luminosity (A) and deforestation (B) in treatment and control group within 10 km of the border. Green dots: treatment, blue squares: control, dashed red line: year of policy implementation. Bottom (C and D): Remote Sensing Imagery. Data from Garg and Shenoy (2021).

Formally, one way to estimate this is the following equation, now with subscripts for clarity:

$$Y_{i,t} = \pi_i + \left(\sum\nolimits_1^N \kappa_t T_t\right) + \left(\sum\nolimits_1^N T_t \times \beta_1 X^A\right) + \left(\sum\nolimits_1^N T_t \times \beta_2 X^B\right) + \sum\nolimits_1^N \tau_t \left(T_t \times D_i\right) + \varepsilon_{i,t} \tag{3}$$

$Y_{i,t}$ denotes the outcome of interest for unit i in time period t, π_i is a unit fixed effect, controlling for time-invariant unit characteristics, $\sum_1^N \kappa_t T_t$ is a vector of time period dummies, controlling for all common shocks and trends. The next two terms (i.e. $\beta_1 X^A$ and $\beta_2 X^B$) control for everything that smoothly changes with the running variable, separately on threshold side and in time period. The term $\sum_1^N \tau_t \left(T_t \times D_i\right)$ quantifies the discontinuity in the outcome variable in each year, and $\varepsilon_{i,t}$ is an error term.

Wang *et al.* (2022) also adopt a difference-in-discontinuities design but in a very different context. Within Switzerland there is a prominent language border that runs through different cantons and regions, with French-speaking communities on one side and German-speaking communities on the other. While sharing the same institutions and economy, the two sides are culturally distinct.[10] In 2014, multiple agri-environmental payments offered to farmers were increased. This provides an interesting natural experiment: There are

10 For example, Filippini and Wekhof (2021). 'The effect of culture on energy efficient vehicle ownership.' *Journal of Environmental Economics and Management*, 105, 102,400. find that the French speaking side has a stronger environmental culture whereas Brown, Henchoz and Spycher (2018). 'Culture and financial literacy: evidence from a within-country language border.' *Journal of Economic Behavior & Organization*, 150, 62–85 and Eugster *et al.* (2017). 'Culture, work attitudes, and job search: evidence from the Swiss language border.' *Journal of the European Economic Association*, 15(5), 1056–100. Find that the German-speaking side has a stronger business culture.

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two culturally distinct groups of farmers and abruptly they are both offered higher agri-environmental payments. Wang *et al*. (2022) find that before the reform, farmers on the French-speaking side of the cultural border systematically enrolled less land into biodiversity conservation, compared to the German-speaking side, everything else (e.g. soil and environmental conditions) being equal. Then, with increased monetary incentives following the 2014 policy reform, the French-speaking farmers enrolled more additional land than the German-speaking farmers, shrinking the border-discontinuity. Another example is the analysis of Baragwanath and Bayi (2020), who study the effect of strengthening indigenous property rights in the Brazilian Amazon on deforestation. For this, they estimate the discontinuity in deforestation at the boundary of indigenous territories, in periods before and after their rights were improved.

3.2. Heterogeneous treatment effects models

In Section 2.2, we already discussed heterogeneity analyses based on aggregating data in different ways and creating subsamples. However, there are also other techniques to do this. For example, Chang and Lin (2020) use quantile regression. Their main research question is the income-effect of a farmland zoning program in Taiwan. After estimating the program's average effect, they analyse whether they can systematically explain how much different farmers are impacted. To investigate whether the effect systematically varies across the farm income distribution they use the quantile treatment effects regression discontinuity model of Frandsen, Frölich and Melly (2012).[11] Estimating separate treatment effects for each 0.1 percentile, Chang and Lin (2020) find that the effect systematically grows with each percentile. The identifying assumptions are the same as for the estimation of the average treatment effect.

Hsu and Shen (2019) propose a test[12] to study whether a policy treatment is (i) beneficial for at least some subpopulations defined by covariate values, (ii) has any impact on at least some subpopulations and (iii) has a heterogeneous impact across subpopulations. This test is similar to the above-discussed analyses of subsample or using interaction effects, but has better statistical properties. Hsu and Shen (2021) also propose a second test to examine whether the estimated average treatment effect or the estimated local average treatment effect has a monotonic relationship with observed individual characteristics. Then, Reguly (2021) proposes a causal supervised machine learning algorithm[13] to uncover treatment effect heterogeneity. This is based on an honest 'regression discontinuity tree', where each leaf of the tree contains an estimated discontinuity conditional on the values of pre-treatment covariates.

11 This can be estimated in Stata using rddqte, available from https://sites.google.com/site/blaisemelly/home/computer-programs/quantile-treatment-effects-in-the-regression-discontinuity-design.

12 This test can be performed in R with codes available from https://sites.google.com/site/innoshu/home/research.

13 This can be estimated in Matlab with codes available from https://github.com/regulyagoston/RD_tree.

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For an introduction to machine learning in agricultural economics, see Storm, Baylis and Heckelei (2020).

3.3. Local polynomial and randomisation approaches

Technically, the different versions of RDDs can be estimated using simple linear regression, or non-parametric local polynomial regression. In principle, the latter approach has several attractive statistical features and it is moreover constantly being improved (e.g. Calonico *et al.*, 2019; Calonico, Cattaneo and Titiunik, 2014). Especially the packages developed and provided by Calonico, Cattaneo, Titiunik and Farrell (Calonico *et al.*, 2019; Calonico, Cattaneo and Titiunik, 2014)—available for Stata, R and Python—offer many useful features, foremost a confidence interval bias-correction that improves empirical coverage, and also automatic checks and corrections for a discrete running variable, mass points and overshooting in the support of the running variable, estimations based on local randomisation instead of the common continuity assumption, and inference and extrapolation methods for designs with multiple cutoffs and scores. All these features and others are extensively covered in Cattaneo, Idrobo and Titiunik (2020).

The approach based on linear regression is especially can especially be biased if a bandwidth is selected that is too wide and observations away from the cutoff affect the estimated discontinuity. This issue can be clearly seen in Figure 5. When using linear regression, getting the bandwidth right is thus particularly important and as a response it is thus common to show that one's estimates are robust to varying the bandwidth. Overall, the majority (but far not all) of applied economics papers is nevertheless based on linear regression. The two main reasons are simplicity and flexibility. It is computationally inexpensive, even with very large datasets and limited computational resources, and can easily be adjusted, augmented, and combined, whereas the non-parametric local polynomial approach makes it difficult to do anything that is not yet included in the ready-to-use packages. However, since the packages are continuously being improved and computational power increases rapidly, chances are that their use will increase in the future,

3.4. Extrapolation away from the threshold

RDDs commonly estimate the 'average treatment effect at the cutoff' (sharp RDD) or the 'complier average treatment effect at the cutoff' (fuzzy RDD). Frequently, however, one is actually interested to learn something about the 'average treatment effect' or the 'average treatment effect on the treated' (Wing and Bello-Gomez, 2018). Above in Section 2.2, we cover some ways how the external validity of regression discontinuity estimates can be probed and assessed, such as estimating the treatment effect derivative (Dong and Lewbel, 2015). Here, we briefly cover approaches that aim to directly extrapolate local estimates to units further away from the cutoff. To the best of our knowledge, these have not yet been applied in agricultural and environmental economics studies but they are tested and ready. The first one is proposed

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by Angrist and Rokkanen (2015)[14] which is a complementary combination of regression discontinuity with matching. This approach requires two assumptions. First, conditional independence and second, common support. The idea is that the RDD partially validates the matching approach and the matching then helps to extrapolate the local estimates to improve external validity. This works because (conditional) independence implies that the expected value of the potential outcomes should not be associated with the assignment variable and this can be tested. If conditional independence cannot be rejected, one can use the matching to extrapolate estimated effects (Angrist and Rokkanen, 2015). Alternatively, Wing and Cook (2013) propose a 'comparative RDD' that requires an untreated comparison group and is based on a common functional form assumption. The idea is that the comparison group shows how the outcome variable would be distributed without the cutoff/treatment. The logic is similar to the parallel trends assumption of difference-in-difference; however, it is not changes over time here but changes along the running variable.

3.5. Spillovers

Spillovers across people and space are a key aspect of agricultural and environmental economics and have been identified as highly relevant for policy evaluation and policy conclusions (Schipmann and Qaim, 2010; Lewis, Barham and Robinson, 2011; Dedehouanou *et al.*, 2018; Stoeffler *et al.*, 2022). In some settings, theory suggests the possibility or even likelihood of local spillovers. For example, when estimating the effect of conservation policies on illegal deforestation there is a chance that deforestation is partly shifted just outside the protection boundary ('leakage'). If panel data is available, this can be assessed in two ways. First, once more one can eyeball the plotted raw data. An example is the study of Burgess, Costa and Olken (2018). They analyse how the deforestation discontinuity at the Brazilian border changed over time in response to Brazil changing its forest policy. They plot all annual discontinuities and one can see that it is in fact the deforestation on the Brazilian side that is changing and not the deforestation on the other side. Local spillovers would create a characteristic pattern in that particularly deforestation right next to the border would go up and the effect would decline with border distance. This is clearly absent in the study of Burgess, Costa and Olken (2018). An econometric test of this is used by Turner, Haughwout and Van Der Klaauw (2014).[15] They analyse how land use regulations affect land values and welfare in the USA, exploiting regulatory differences across municipalities (using municipality boundaries as thresholds). To quantify the effect of one municipality's regulation on the land prices in the neighboring municipalities, they make the assumption that this spillover is local. This implies that by comparing price changes for peripheral and interior land one can learn about spillovers, as long as one can control for other differences between these regions.

14 Stata do-files available here: https://economics.mit.edu/faculty/angrist/data1/data/angrist6.

15 Stata do-files available here: https://www.econometricsociety.org/content/supplement-land-use-regulation-and-welfare.

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3.6. Using modelled agricultural and environmental data

Often, measurements of variables of interests are not available and researchers need to either adapt the questions they are asking or rely on whatever proxies are available. An illustrative example is the study of Wuepper, Borrelli and Finger (2020a), which estimates international border discontinuities in soil erosion. Soil erosion cannot fully be captured with remote sensing and even if all national governments around the world would send out soil scientists to measure soil erosion in the field, we would be worried that national statistics would be affected by measurement differences between the countries. To overcome this data gap and to test the role of countries and their institutions and policies on soil erosion, Wuepper, Borrelli and Finger (2020a) rely on globally modelled soil erosion data from Borrelli *et al*. (2017).

Another variable that is also not available is the counterfactual. A central assumption in the study is that without countries there would not be soil erosion discontinuities in the locations where there are borders now. As discussed above, researchers commonly analyse environmental covariates to test such an assumption. Wuepper, Borrelli and Finger (2020a), however, directly model the natural rate of soil erosion to directly test whether, and if so where, the natural rate of soil erosion would abruptly change at country borders. One example for this is shown in Figure 1. The modelling of the counterfactual soil erosion is done exactly like the modelling of the actual soil erosion, except it is modelled without human influences, which is the counterfactual of interest in this particular study. Bastin *et al*. (2019) provide a global map of the *natural* tree-cover and using global environmental data one can fill-in the most likely remaining land-cover. This way, one can globally model the counterfactual outcome without humans, e.g. the natural rate of soil erosion.[16] This can be used to test assumptions, partition subsamples, and to include it as a covariate in specifications.

4. Emerging (data) opportunities

There are many emerging opportunities in agricultural, environmental and related economics fields. These can largely also improve the potential for agricultural and environmental policy evaluation. Here, we particularly want to emphasise emerging opportunities that arise from new data sources. This includes the ever-growing availability of high-resolution geospatial measurements (**#1**), data generated with machine learning (**#2**), and simply data that contains policy relevant thresholds that have not been exploited yet (**#3**).

4.1. #1 Geospatial data and remote sensing opportunities

Geospatial platforms such as Google Earth Engine (2022) make it easy and inexpensive for researchers to collect unprecedented amounts and details of

16 Such maps already exists for forests, soil erosion and above- and below-ground carbon, available at https://esdac.jrc.ec.europa.eu/content/land-degradation-debt.

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data from all over the world (Gorelick *et al.*, 2017). This includes, e.g. environmental variables such as measures of biodiversity decline, land degradation, or water pollution, as well as socio-economic variables such as measures of crop yields (e.g. using vegetation indices) or incomes (e.g. using nighttime lights). Once we have learned about the existence of viable thresholds that affect some treatment of interest, it is often possible to directly download all the required data that is needed, for example remotely sensed agricultural production (Burke *et al.*, 2021; Lobell *et al.*, 2019), and estimate whether there is a discontinuity that identifies a causal effect.

Remote sensing data implies massive potentials for economics in general, but especially for agricultural and environmental economics. See Donaldson and Storeygard (2016) for general discussion and Jain (2020) and Alix-Garcia and Millimet (2021) for methodological discussions. Overall, the growing availability of agricultural and environmental data is especially valuable to agricultural economists who are commonly trained both in agricultural and related sciences (e.g. plant nutrition, soil science) as well as in economic theory and analysis (e.g. optimal control, econometrics). Much of this is urgently needed to contribute to research on sustainable development and can enable highly policy relevant studies.

4.2. #2 Machine learning to generate variables

A constraint for many empirical studies is data availability. When evaluating the effect of agri-environmental schemes, the variables of interest might be changes in local biodiversity, soil erosion, or greenhouse gas emissions, but these might not be available as variables in the data and thus researchers must rely on proxies, such as area shares of different crops, land-cover changes, etc. In Section 3.6 above, we already discussed the combination of RDDs and modelling agricultural and environmental data. The cited studies, however, just mark the beginning. It is possible to use machine learning, e.g. random forest algorithms, to generate new variables, even directly in Google Earth Engine (2022). Burke *et al.* (2021) discuss a variety of possible avenues for this. For example, one could predict high-resolution agricultural profits, or annual rates of soil erosion at the field level, or many other variables that would otherwise be difficult to obtain at the same scope and scale. For a discussion of potential issues with measurement error, see Section 2.2. issue #5 'Measurement Error' above.

4.3. #3 Neglected datasets

A perhaps less impressive, but especially efficient source of new data is data which exists already but researchers are unaware of it or have not yet used it for other reasons. It is not uncommon that policy makers use sharp thresholds to determine either who is affected by a policy change or how much different segments of the population/regions are affected. In many countries, agricultural policy (e.g. subsidies, regulations, taxes) differs by farm size and the thresholds are discrete (e.g. below X hectares, or above Y standardised

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labour units).[17] Another example are policy boundaries that are unique in the sense that they do not entirely coincide with other boundaries, e.g. the agricultural areas of a country might be divided into discrete growing zones, or a region might be categorised into different biomes, even though the actual change in conditions is continuous. In such instances, one part of a forest might be strictly protected because it is 'natural' and the other part is free for use because it is 'managed', even though close to the threshold both sides are equally 'natural'/'managed', or some farms are categorised to be in a 'high-rainfall area' and others are categorised to be in the 'low rainfall area', but close to the threshold they actually receive the same amount of rainfall. As a final example, insurances often use sharp thresholds that trigger payouts. Actively searching for such data and talking with stakeholders will likely reveal a large number of feasible datasets that contain the answers to important policy questions.

5. Conclusion

The RDD in its various forms is becoming increasingly popular in agricultural and environmental economics. Its main 'selling point' is commonly the transparency with which the validity of key identifying assumptions can be probed. This is key also for the assessment of agricultural and environmental policies. Important challenges for applied researchers include that RDDs can be specified in many ways, with sometimes obvious and sometimes subtle differences and implications. Moreover, the *ability* to probe all key identifying assumptions basically translates into a norm that they *should* all be probed—and again, often there are multiple options to do so. Not all policy questions related to agricultural and environmental issues can be addressed using RDDs. However, we here show that there still remains much unexploited potential to employ RDDs for research and policy.

In this review, we provide many empirical examples from agricultural and environmental economics and policy evaluation to showcase the range of questions that can be answered and to inspire research to follow some of these examples in new directions. Secondly, we have used many of these examples to illustrate the various ways one can probe the key identifying assumptions of the different versions of the approach. The identification threats that we mainly discussed included misleading distributions (e.g. non-linear functions that might produce spurious linear discontinuities), endogenous thresholds (e.g. there exists discontinuously distributed confounders), endogenous sorting (e.g. some units cross the threshold to receive the treatment), missing units

17 For example, in the European Union, farms below 10 ha need less crop diversity than farms above 10, and the same is true again for farms below and above 30. In addition, only farms above 15 ha need to create 'Ecological Focus Areas'. Wuepper, Wimmer and Sauer (2020c). 'Is small family farming more environmentally sustainable? Evidence from a spatial regression discontinuity design in Germany.' *Land use policy*, 90, 104,360. And in Switzerland, farms above one standardised labour unit are granted a lower farm acquisition price when a family member takes over the business compared to farms below one standardised labour unit. Huber, Flury and Finger (2015). 'Factors affecting farm growth intentions of family farms in mountain regions: empirical evidence for central Switzerland.' Ibid.47, 188–97.

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(e.g. insufficient data close to the threshold), measurement error (classical and non-classical), imperfect compliance (treatment assignment only increases *probability* of treatment), and last but not least, external validity (causal effect at the threshold is a special case that does not generalise).

We show that various extensions can even further improve policy evaluation. These include the difference-in-discontinuities design. This combination of RDD and difference-in-differences adds the assumption of parallel trends, which, simultaneously becomes more likely to hold than in a difference-in-differences framework, because of the focus on discontinuities and units near the threshold. Thus, this is highly suited to identify implications of new policies and policy changes over time. Moreover, we discussed spillovers and ways to extrapolate findings away from the threshold, as well as how to integrate bio-physical modelling, e.g. of crop yields or environmental issues.

Regarding emerging opportunities, we identify increasing collaborations across disciplines. This includes the integration of spatial data from remote sensing, using machine learning to generate useful new data, and discovering policy rules based on sharp thresholds.

Our review shows that RDDs are becoming a key element in the toolbox of agricultural and environmental economists and has become a key approach for agricultural and environmental policy evaluation. We show that methodological extensions and opportunities arising from new data sources even increase the policy relevance of RDDs. The two main challenges going forward are to better utilise the insights from RDD in actual policy making, e.g. when reforming the Common Agricultural Policy, and to not only advance the frontier of the method but also to move the average closer to the existing frontier. This requires more awareness of best practices, opportunities and pitfalls. Also providing (better access to) data sources useful for RDDs (e.g. satellite data, geocoded survey and farm-level data) can be a task for policy and administration. A clear benefit of the method is that it helps communication because many aspects can be visualised, and it is well suited for inter- and transdisciplinary research, across scales. All this is urgently needed to address our current sustainability challenges and is what agricultural and environmental economists are particularly well trained to do. Strong causal identification is commonly a requirement for reliable and convincing policy recommendations and RDDs excel at this.

Acknowledgements

We thank Salvatore Di Falco, Jennifer Alix-Garcia and two anonymous reviewers for their kind and constructive feedback and suggestions.

References

Abadie, A. and Cattaneo, M. D. (2018). Econometric methods for program evaluation. *Annual Review of Economics* 10: 465–503. 10.1146/annurev-economics-080217-053402.

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Ali, D. A., Deininger, K. and Goldstein, M. (2014). Environmental and gender impacts of land tenure regularization in Africa: pilot evidence from Rwanda. *Journal of Development Economics* 110: 262–275. 10.1016/j.jdeveco.2013.12.009.

Alix-Garcia, J., McIntosh, C., Sims, K. R. E. and Welch, J. R. (2013). The ecological footprint of poverty alleviation: evidence from Mexico's Oportunidades Program. *Review of Economics and Statistics* 95(2): 417–435. 10.1162/REST_a_00349.

Alix-Garcia, J. and Millimet, D. L. (2021). Remotely incorrect? Accounting for nonclassical measurement in satellite data on deforestation. Working Paper Oregon State University, 28 January 2021.

Alix-Garcia, J. M., Sims, K. R. E., Orozco-Olvera, V. H., Costica, L. E., Fernández Medina, J. D. and Romo Monroy, S. (2018). Payments for environmental services supported social capital while increasing land management. *Proceedings of the National Academy of Sciences* 115(27): 7016–7021. 10.1073/pnas.1720873115.

Angrist, J. D., Imbens, G. W. and Rubin, D. B. (1996). Identification of causal effects using instrumental variables. *Journal of the American Statistical Association* 91(434): 444–455. 10.1080/01621459.1996.10476902.

Angrist, J. D. and Pischke, J.-S. (2010). The credibility revolution in empirical economics: how better research design is taking the con out of econometrics. *Journal of Economic Perspectives* 24(2): 3–30. 10.1257/jep.24.2.3.

Angrist, J. D. and Rokkanen, M. (2015). Wanna get away? Regression discontinuity estimation of exam school effects away from the cutoff. *Journal of the American Statistical Association* 110(512): 1331–1344. 10.1080/01621459.2015.1012259.

Appau, S., Churchill, S. A., Smyth, R. and Trinh, T.-A. (2021). The long-term impact of the Vietnam War on agricultural productivity. *World Development* 146: 105613. 10.1016/j.worlddev.2021.105613.

Asher, S. and Novosad, P. (2017). Politics and local economic growth: evidence from India. *American Economic Journal: Applied Economics* 9(1): 229–273. 10.1257/app.20150512.

Asher, S. and Novosad, P. (2020). Rural roads and local economic development. *American Economic Review* 110(3): 797–823. 10.1257/aer.20180268.

Ayres, A. B., Meng, K. C., and Plantinga, A. J. (2021). Do environmental markets improve on open access? Evidence from California groundwater rights. *Journal of Political Economy* 129(10): 2817–2860.

Bagues, M. and Campa, P. (2021). Can gender quotas in candidate lists empower women? Evidence from a regression discontinuity design. *Journal of Public Economics* 194: 104315. 10.1016/j.jpubeco.2020.104315.

Baragwanath, K. and Bayi, E. (2020). Collective property rights reduce deforestation in the Brazilian Amazon. *Proceedings of the National Academy of Sciences* 117(34): 20495–20502. 10.1073/pnas.1917874117.

Bastin, J.-F., Finegold, Y., Garcia, C., Mollicone, D., Rezende, M., Routh, D., Zohner, C. M. and Crowther, T. W. (2019). The global tree restoration potential. *Science* 365(6448): 76–79. 10.1126/science.aax0848.

Behaghel, L., Macours, K. and Subervie, J. (2019). How can randomised controlled trials help improve the design of the common agricultural policy? *European Review of Agricultural Economics* 46(3): 473–493. 10.1093/erae/jbz021.

Bonilla-Mejía, L. and Higuera-Mendieta, I. (2019). Protected areas under weak institutions: evidence from Colombia. *World Development* 122: 585–596. 10.1016/j.worlddev.2019.06.019.

Borrelli, P., Robinson, D. A., Fleischer, L. R., Lugato, E., Ballabio, C., Alewell, C., Meusburger, K., Modugno, S., Schütt, B. and Ferro, V. (2017). An assessment of

Downloaded from https://academic.oup.com/erae/article/50/1/1/6749527 by guest on 28 April 2025

the global impact of twenty-first century land use change on soil erosion. *Nature Communications* 8(1): 2013. 10.1038/s41467-017-02142-7.

Brodeur, A., Cook, N. and Heyes, A. (2020). Methods matter: P-hacking and publication bias in causal analysis in economics. *American Economic Review* 110(11): 3634–3660. 10.1257/aer.20190687.

Brown, M., Henchoz, C. and Spycher, T. (2018). Culture and financial literacy: evidence from a within-country language border. *Journal of Economic Behavior & Organization* 150: 62–85. 10.1016/j.jebo.2018.03.011.

Burgess, R., Costa, F. J. M. and Olken, B. A. (2018). Wilderness conservation and the reach of the state: evidence from national borders in the Amazon. National Bureau of Economic Research.

Burke, M., Driscoll, A., Lobell, D. B. and Ermon, S. (2021). Using satellite imagery to understand and promote sustainable development. *Science* 371(6535): 1–13. 10.1126/science.abe8628.

Butts, K. (2021). Geographic difference-in-discontinuities. *Applied Economics Letters* 1–5. 10.1080/13504851.2021.2005236.

Calonico, S., Cattaneo, M. D. and Farrell, M. H. (2020). Optimal bandwidth choice for robust bias-corrected inference in regression discontinuity designs. *The Econometrics Journal* 23(2): 192–210. 10.1093/ectj/utz022.

Calonico, S., Cattaneo, M. D., Farrell, M. H. and Titiunik, R. (2019). Regression discontinuity designs using covariates. *The Review of Economics and Statistics* 101(3): 442–451. 10.1162/rest_a_00760.

Calonico, S., Cattaneo, M. D. and Titiunik, R. (2014). Robust nonparametric confidence intervals for regression-discontinuity designs. *Econometrica* 82(6): 2295–2326. 10.3982/ECTA11757.

Canay, I. A. and Kamat, V. (2018). Approximate permutation tests and induced order statistics in the regression discontinuity design. *The Review of Economic Studies* 85(3): 1577–1608. 10.1093/restud/rdx062.

Cattaneo, M. D., Idrobo, N. and Titiunik, R. (2020). *A Practical Introduction to Regression Discontinuity Designs: Foundations and Extensions*. Cambridge, UK: Cambridge University Press.

Cattaneo, M. D. and Titiunik, R. (2021). Regression discontinuity designs. *arXiv preprint arXiv:2108.09400*.

Cattaneo, M. D. and Vazquez-Bare, G. (2016). The choice of neighborhood in regression discontinuity designs. *Observational Studies* 2(134): A146. 10.1353/obs.2017.0002.

Chang, H.-H. and Lin, T.-C. (2016). Does the minimum lot size program affect farmland values? Empirical evidence using administrative data and regression discontinuity design in Taiwan. *American Journal of Agricultural Economics* 98(3): 785–801. 10.1093/ajae/aav064.

Chang, H.-H. and Lin, T.-C. (2020). Does a farmland zoning program impact farm income: empirical evidence from farm households in Taiwan. *European Review of Agricultural Economics* 47(5): 1621–1643. 10.1093/erae/jbaa002.

Chaplin, D. D., Cook, T. D., Zurovac, J., Coopersmith, J. S., Finucane, M. M., Vollmer, L. N. and Morris, R. E. (2018). The internal and external validity of the regression discontinuity design: a meta-analysis of 15 within-study comparisons. *Journal of Policy Analysis and Management* 37(2): 403–429. 10.1002/pam.22051.

Chen, Y., Shi, S. and Tang, Y. (2019). Valuing the urban hukou in China: evidence from a regression discontinuity design for housing prices. *Journal of Development Economics* 141: 102381. 10.1016/j.jdeveco.2019.102381.

Downloaded from https://academic.oup.com/erae/article/50/1/1/6749527 by guest on 28 April 2025

Colen, L., Gomez Y Paloma, S., Latacz-Lohmann, U., Lefebvre, M., Préget, R. and Thoyer, S. (2016). Economic experiments as a tool for agricultural policy evaluation: insights from the European CAP. *Canadian Journal of Agricultural Economics/Revue Canadienne D'agroeconomie* 64(4): 667–694. 10.1111/cjag.12107.

Cunningham, S. (2021). *Causal Inference*. New Haven, USA: Yale University Press.

Dedehouanou, S. F. A., Araar, A., Ousseini, A., Laouali Harouna, A. and Jabir, M. (2018). Spillovers from off-farm self-employment opportunities in Rural Niger. *World Development* 105: 428–442. 10.1016/j.worlddev.2017.12.005.

De Janvry, A., Ritchie, E. R. and Sadoulet, E. (2016). Weather index insurance and shock coping: evidence from Mexico's Cadena Program. *World Bank Policy Research Working Paper*, (7715).

Donaldson, D. and Storeygard, A. (2016). The view from above: applications of satellite data in economics. *Journal of Economic Perspectives* 30(4): 171–198. 10.1257/jep.30.4.171.

Dong, Y. and Lewbel, A. (2015). Identifying the effect of changing the policy threshold in regression discontinuity models. *Review of Economics and Statistics* 97(5): 1081–1092. 10.1162/REST_a_00510.

Esposti, R. and Sotte, F. (2013). Evaluating the effectiveness of agricultural and rural policies: an introduction. *European Review of Agricultural Economics* 40(4): 535–539. 10.1093/erae/jbt014.

Eugster, B., Lalive, R., Steinhauer, A. and Zweimüller, J. (2017). Culture, work attitudes, and job search: evidence from the Swiss language border. *Journal of the European Economic Association* 15(5): 1056–1100. 10.1093/jeea/jvw024.

Filippini, M. and Wekhof, T. (2021). The effect of culture on energy efficient vehicle ownership. *Journal of Environmental Economics and Management* 105: 102400. 10.1016/j.jeem.2020.102400.

Finger, R. and El Benni, N. (2021). Farm income in European agriculture: new perspectives on measurement and implications for policy evaluation. *European Review of Agricultural Economics* 48(2): 253–265. 10.1093/erae/jbab011.

Frandsen, B. R., Frölich, M. and Melly, B. (2012). Quantile treatment effects in the regression discontinuity design. *Journal of Econometrics* 168(2): 382–395. 10.1016/j.jeconom.2012.02.004.

Garg, T. and Shenoy, A. (2021). The ecological impact of place-based economic policies. *American Journal of Agricultural Economics* 103(4): 1239–1250. 10.1111/ajae.12193.

Ge, M., Edwards, E. C. and Akhundjanov, S. B. (2020). Irrigation investment on an American Indian reservation. *American Journal of Agricultural Economics* 102(4): 1083–1104. 10.1002/ajae.12077.

Gelman, A. and Imbens, G. (2019). Why high-order polynomials should not be used in regression discontinuity designs. *Journal of Business and Economic Statistics* 37(3): 447–456. 10.1080/07350015.2017.1366909.

Gleason, P., Resch, A. and Berk, J. (2018). RD or not RD: using experimental studies to assess the performance of the regression discontinuity approach. *Evaluation Review* 42(1): 3–33. 10.1177/0193841X18787267.

Google Earth Engine. (2022). https://earthengine.google.com/.

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D. and Moore, R. (2017). Google Earth Engine: planetary-scale geospatial analysis for everyone. *Remote Sensing of Environment* 202: 18–27. 10.1016/j.rse.2017.06.031.

Hansen, M. C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S., Tyukavina, A., Thau, D., Stehman, S. V., Goetz, S. J. and Loveland, T. R. (2013). High-resolution global

Downloaded from https://academic.oup.com/erae/article/50/1/1/6749527 by guest on 28 April 2025

maps of 21st-century forest cover change. *Science* 342(6160): 850–853. 10.1126/science.1244693.

Hsu, Y.-C. and Shen, S. (2019). Testing treatment effect heterogeneity in regression discontinuity designs. *Journal of Econometrics* 208(2): 468–486. 10.1016/j.jeconom.2018.10.004.

Hsu, Y. and Shen, S. (2021). Testing monotonicity of conditional treatment effects under regression discontinuity designs. *Journal of Applied Econometrics* 36(3): 346–366. 10.1002/jae.2805.

Huber, R., Flury, C. and Finger, R. (2015). Factors affecting farm growth intentions of family farms in mountain regions: empirical evidence for central Switzerland. *Land Use Policy* 47: 188–197. 10.1016/j.landusepol.2015.04.006.

Huntington-Klein, N. (2021). *The Effect: An Introduction to Research Design and Causality*. London, UK: Chapman and Hall/CRC.

Imbens, G. and Lemieux, T. (2008). Regression discontinuity designs: a guide to practice. *Journal of Econometrics* 142(2): 615–635. 10.1016/j.jeconom.2007.05.001.

Jain, M. (2020). The benefits and pitfalls of using satellite data for causal inference. *Review of Environmental Economics and Policy* 14(1): 157–169. 10.1093/reep/rez023.

Jones, M., Kondylis, F., Loeser, J. and Magruder, J. (2022). Factor market failures and the adoption of irrigation in rwanda. *American Economic Review* 112(7): 2316–2352.

Keele, L. J. and Titiunik, R. (2014). Geographic boundaries as regression discontinuities. *Political Analysis* 23(1): 127–155. 10.1093/pan/mpu014.

Khanna, G. and Zimmermann, L. (2017). Guns and butter? Fighting violence with the promise of development. *Journal of Development Economics* 124: 120–141. 10.1016/j.jdeveco.2016.09.006.

Lee, D. S. and Lemieux, T. (2010). Regression discontinuity designs in economics. *Journal of Economic Literature* 48(2): 281–355. 10.1257/jel.48.2.281.

Lewis, D. J., Barham, B. L. and Robinson, B. (2011). Are there spatial spillovers in the adoption of clean technology? The case of organic dairy farming. *Land Economics* 87(2): 250–267. 10.3368/le.87.2.250.

Lobell, D. B., Azzari, G., Burke, M., Gourlay, S., Jin, Z., Kilic, T. and Murray, S. (2019). Eyes in the sky, boots on the ground: assessing satellite- and ground-based approaches to crop yield measurement and analysis. *American Journal of Agricultural Economics* 102: 202–219. 10.1093/ajae/aaz051.

MacPherson, C. and Sterck, O. (2021). Empowering refugees through cash and agriculture: a regression discontinuity design. *Journal of Development Economics* 149: 102614. 10.1016/j.jdeveco.2020.102614.

McCrary, J. (2008). Manipulation of the running variable in the regression discontinuity design: a density test. *Journal of Econometrics* 142(2): 698–714. 10.1016/j.jeconom.2007.05.005.

Nguyen, T. C. and Rieger, M. (2017). Community-driven development and social capital: evidence from Morocco. *World Development* 91: 28–52. 10.1016/j.worlddev.2016.10.013.

Noack, F., Larsen, A., Kamp, J. and Levers, C. (2022). A bird's eye view of farm size and biodiversity: the ecological legacy of the iron curtain. *American Journal of Agricultural Economics* 104(4): 1460–1484. 10.1111/ajae.12274.

Nordin, M. and Lovén, I. (2020). Is the setting up aid mitigating the generational renewal problem in farming? *European Review of Agricultural Economics* 47(5): 1697–1715. 10.1093/erae/jbaa006.

Downloaded from https://academic.oup.com/erae/article/50/1/1/6749527 by guest on 28 April 2025

Pan, Y., Smith, S. C. and Sulaiman, M. (2018). Agricultural extension and technology adoption for food security: evidence from Uganda. *American Journal of Agricultural Economics* 100(4): 1012–1031. 10.1093/ajae/aay012.

Potapov, P., Turubanova, S., Hansen, M., Tyukavina, A., Zalles, V., Khan, A., Song, X.-P., Pickens, A., Shen, Q. and Cortez, J. (2021). Global maps of cropland extent and change show accelerated cropland expansion in the twenty-first century. *Nature Food* 1–10. 10.1038/s43016-021-00429-z.

Reguly, Á. (2021). Heterogeneous treatment effects in regression discontinuity designs. *arXiv preprint arXiv:2106*.11640.

Rodriguez, O., Vrachiolo, M., Wuepper, D. and Sauer, J. (2022). World heritage status and farmers' income: evidence from a regression discontinuity design in Colombia.

Ruan, J., Cai, Q. and Jin, S. (2021). Impact of COVID-19 and nationwide lockdowns on vegetable prices: evidence from wholesale markets in China. *American Journal of Agricultural Economics* 103(5): 1574–1594. 10.1111/ajae.12211.

Salazar, L., Aramburu, J., Agurto, M., Maffioli, A. and Fahsbender, J. (2020). Sweeping the flies away: evidence from a fruit fly eradication program. *European Review of Agricultural Economics* 47(5): 1920–1962. 10.1093/erae/jbaa015.

Salman, M., Long, X., Wang, G. and Zha, D. (2022). Paris climate agreement and global environmental efficiency: new evidence from fuzzy regression discontinuity design. *Energy Policy* 168: 113128. 10.1016/j.enpol.2022.113128.

Schipmann, C. and Qaim, M. (2010). Spillovers from modern supply chains to traditional markets: product innovation and adoption by smallholders. *Agricultural Economics* 41(3-4): 361–371. 10.1111/j.1574-0862.2010.00438.x.

Shenoy, A. (2018). Regional development through place-based policies: evidence from a spatial discontinuity. *Journal of Development Economics* 130: 173–189. 10.1016/j.jdeveco.2017.10.001.

Shi, X. (2022). Land reallocation responses to China's new rural pension scheme: evidence from a regression discontinuity design. *Journal of Rural Studies* 92: 35–49. 10.1016/j.jrurstud.2022.03.018.

Stoeffler, Q., Carter, M., Guirkinger, C. and Gelade, W. (2022). The spillover impact of index insurance on agricultural investment by cotton farmers in Burkina Faso. *The World Bank Economic Review* 36(1): 114–140. 10.1093/wber/lhab011.

Storm, H., Baylis, K. and Heckelei, T. (2020). Machine learning in agricultural and applied economics. *European Review of Agricultural Economics* 47(3): 849–892. 10.1093/erae/jbz033.

Turner, M. A., Haughwout, A. and Van Der Klaauw, W. (2014). Land use regulation and welfare. *Econometrica* 82(4): 1341–1403. 10.3982/ECTA9823.

Wang, Y., Schaub, S., Wuepper, D. and Finger, R. (2022). Culture and agricultural biodiversity conservation. *Working paper*, available at SSRN: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4043727.

Wing, C. and Bello-Gomez, R. A. (2018). Regression discontinuity and beyond: options for studying external validity in an internally valid design. *American Journal of Evaluation* 39(1): 91–108. 10.1177/1098214017736155.

Wing, C. and Cook, T. D. (2013). Strengthening the regression discontinuity design using additional design elements: a within-study comparison. *Journal of Policy Analysis and Management* 32(4): 853–877. 10.1002/pam.21721.

Wuepper, D., Borrelli, P. and Finger, R. (2020a). Countries and the global rate of soil erosion. *Nature Sustainability* 3(1): 51–55. 10.1038/s41893-019-0438-4.

Downloaded from https://academic.oup.com/erae/article/50/1/1/6749527 by guest on 28 April 2025

Wuepper, D., Le Clech, S., Zilberman, D., Mueller, N. and Finger, R. (2020b). Countries influence the trade-off between crop yields and nitrogen pollution. *Nature Food* 1(11): 713–719. 10.1038/s43016-020-00185-6.

Wuepper, D., Wimmer, S. and Sauer, J. (2020c). Is small family farming more environmentally sustainable? Evidence from a spatial regression discontinuity design in Germany. *Land Use Policy* 90: 104360. 10.1016/j.landusepol.2019.104360.

Wuepper, D., Borrelli, P., Panagos, P., Lauber, T., Crowther, T., Thomas, A. and Robinson, D. A. (2021). A 'debt' based approach to land degradation as an indicator of global change. *Global Change Biology* 27(21): 5407–5410. 10.1111/gcb.15830.

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